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AI Based Traffic Congestion Monitoring and Alert System

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***Abstract*—Traffic congestion is a severe problem in urban cities due to population growth, which leads to an increase in the number of vehicles, creating high demand during peak hours, road accidents, and unstable climate conditions. Conventional traffic management systems are based on either manual or reactive analysis that often results in late congestion detection and ineffective response strategies. The proposed study is an AI-Based Traffic Congestion Monitoring and Alert System, which functions as an intelligent and assistive decision supporting system. The system uses a multi-agent architecture consisting of Traffic Data Analysis Agent, Congestion Trend Detection Agent, and Traffic-Alert Advisory Assistant that will constantly process real-time and previous traffic data, covering factors such as speed of vehicles, traffic density, road occupancy, and any traffic incidents. The implementation of the system becomes possible with the help of LangFlow in combination with IBM Granite foundation model and Retrieval- Augmented Generation (RAG) that enables the integration of traffic management rules, congestion levels, and reliable transport data. Experimental results demonstrate an early-stage congestion detection, enhanced situational awareness, and proactive support for traffic management. It is a non-enforcement system that gives interpretable alerts and route level advice to help traffic authorities make informed decisions.**

Keywords: Artificial Intelligence, Traffic Congestion Monitoring, Multi-Agent System, Retrieval-Augmented Generation, Intelligent Transportation Systems

I. INTRODUCTION

Traffic congestion is one of the challenges that have persisted over a long time in the contemporary cities due to the rapid urbanization and steady increase in vehicular volume. Congestion not only adds to the time and fuel used in transportation but also plays a major part in environmental pollution, financial loss and stress among commuters. Traffic in the urban environment is very dynamic and changes with the interplay of several factors including vehicle density, peak-hour demand, road geometry, accidents, construction works, weather changes and special events. Though the majority of metropolitan areas implement traffic management systems which gather massive amounts of real-time and historical traffic data by means of sensors, cameras, GPS- enabled cars, and control centers, the useful application of this data is minimal. Old ways of watching traffic usually depend on people doing it manually or checking what happened after a crash. Due to this, these methods aren't very good at spotting traffic jams before they actually start. Due to this, the traffic authorities are often left reacting to traffic congestion when it has already grown out of proportion and hence, traffic management is always reactive instead of proactive. Recent developments in the field of artificial intelligence introduced new opportunities in the world of intelligent transportation systems because they provided openings to conduct continuous data analysis, identify patterns, and make predictions.

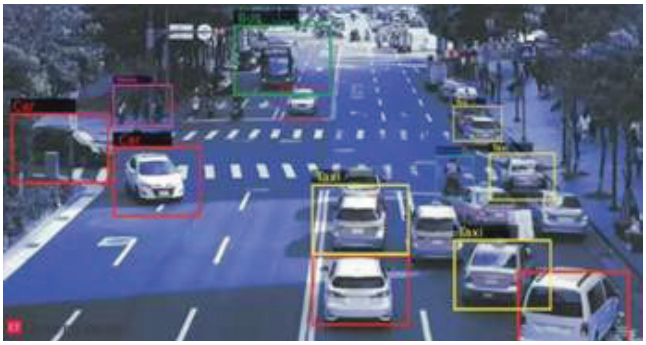


Fig. 1. AI-Based Real-Time Vehicle Detection and Traffic Density Analysis Using Computer Vision

AI-oriented systems can handle heterogeneous traffic data streams and detect subtle patterns to suggest abnormal slowdowns or developing initial traffic before serious congestion sets in. Nevertheless, most of the current AI-based solutions focus entirely on making accurate guesses, but they fail in three practical areas: they are not interpretable, they can't adjust instantly to new data (real-time adaptation), and they don't follow standard traffic management rules. Furthermore, they usually do not get adopted operationally due to the lack of human-friendly, supporting alert systems. In this regard, the current research proposes an AI-Based Traffic Congestion Monitoring and Alert System, which uses a multi-agent assistive design to improve situational awareness and promote active traffic control. The suggested system involves the utilization of special agents to analyse traffic data and detect trends in congestion and create alerts, which will allow tracking the short-term and long-term trends in traffic simultaneously. The system uses LangFlow and the IBM Granite foundation model along with the Retrieval-Augmented Generation using trusted transportation references to guarantee the provision of context-sensitive, interpretable and guideline- conformant alerts. The system is explicitly a non- enforcement decision-support tool, which is intended to

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help traffic authorities in the early detection of congestion and strategize a response plan, thus enhancing the efficiency of urban mobility and its resilience.

A. Scope of the Study

This research is confined to the design and testing of an AI-based, assistive traffic congestion monitoring and alerts platform, which can be used in proactive traffic management in cities. The analysis concentrates on the real-time and previous traffic records (including the vehicle velocity, the number of vehicles, the occupancy of the roads, and the reports of accidents) to allow identifying the trends of congestion at the initial stage. The suggested multiagent system focuses on traffic data analysis, congestion trend detection and alerting, based on the LangFlow system, which runs on the IBM Granite foundation model. Retrieval-Augmented Generation is utilized in order to include the traffic management guidelines, congestion threshold standards and trusted transportation references in order to provide context-relevant decision support. The system is tested under a simulated or controlled environment and does not require enforcement, automated control or real-time actuation of a signal. Outside variables like DAM (digital asset management), behaviour modelling, autonomous vehicle coordination and wide-scale implementation of infrastructure are deemed beyond the current study.

B. Need of the Study

The growing complexity of the urban traffic systems and the swift rise in the number of vehicles have increased the issue of traffic congestion, rendering the conventional methods of traffic monitoring and control incompetent. The current traffic management systems are usually based on manual control, predefined rates, or slow processing of traffic information making it ineffective in notifying about traffic congestion early. This responsive character leads to inefficient traffic control policies, increased travel delays, more fuel wastage, and high rates of environmental pollution. It is urgently required to have a smart system capable of constantly processing real-time and historical traffic data in large amounts and determining when a situation may become congested. This requirement is satisfied in this study by suggesting an assistive and AI-based congestion monitoring and alert system that improves situational awareness among traffic authorities.

II. Background of Urban Traffic Congestion

Increased congestion in the urban traffic has become a major problem in the fast-developing cities because of the high rate of urbanization, high population and the growing usage of the personal vehicle as a mode of day-to-day transport. Urban population growth is increasing at the rate where the current road infrastructure is unable to hold the traffic that it is usually presented with, hence common congestion as people drive during peak hours. Other reasons that contribute to the level of congestion include uneven distribution of employment centres, poor public transport network, road network congestion, uncertain traffic flow as well as poor traffic control systems. Moreover, unplanned urban growth, regular road repair works, accidents, and weather-related delays bring about unexpected changes in the traffic flows that cannot be controlled with the standard methods.



Fig. 2. AI and IoT-Enabled Smart City Traffic Monitoring and Congestion Management Framework

The socio-economic and environmental impacts of congestion are far-reaching in nature causing more increased travel duration, decreased productivity, increased fuel consumption, along with greater emission of greenhouse gases and air pollution. Chronic congestion also led to stress among the commuters, road safety threats, as well as poor quality of life in the cities. The current cities provide enormous volumes of traffic information using sensors, surveillance cameras, GPS-equipped vehicles, and mobile applications; still, the successful interpretation of this information is a significant challenge.



Fig. 3. Urban Traffic Intersection with Intelligent Traffic Signal Control

Traditional traffic lights and control systems usually follow set schedules or rely on old reports. Since they stick to these rigid rules, they can't react quickly when road conditions become dynamic. Consequently, road officials often have to embrace reactive solutions when congestion has already been created. The context and the causal factors of the urban traffic jam are, thus, relevant to the formulation of smart, evidence-based measures that may assist in detecting problems early, making decisions, and dealing with traffic dynamics instantly in multifaceted cities.

III. Impact of Congestion on Urban Mobility and Economy

The effects of traffic congestion are multidimensional and far-reaching when it comes to mobility in the cities and the overall economic performance of the cities. Mobility wise, congestion will cause an increase in travel time, unreliability of journeys and travel time variability, making the process of commuting on a daily basis stressful and inefficient to people. Delays in the operation of the public transportation networks based on road networks with congestion also make them less

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attractive and reliable, and stimulate the use of personal vehicles, which in turn triggers the congestion cycle. Freight transportation and emergency services have slow response times that may be of serious safety and logistical importance. Congestion is costly economically to urban areas in terms of lost productivity, consumption of fuel and high cost of vehicle operation. Extensive idling and stop and start traffic will lead to over consumption of fuel and an accelerated degradation of vehicles which will inflate the cost of maintenance of both the individuals and the businesses. At the macroeconomic level, congestion leads to negative impacts on the supply chain efficiency in the form of postponement of the goods flow, escalation of the logistics expenses, and low competitiveness of the urban economies. The environmental externalities, such as increased carbon dioxide and nitrogen oxides, are forms of pollution that led to increased public health spending and environmental degradation in the long run. These external expenses also weigh down on an urban economy and lower the standard of living. The cumulative effect of congestion is therefore not only in the inefficiency of transportation, but also in economic growth, urban sustainability and social well being.

IV. Role of Artificial Intelligence in Intelligent Transportation Systems

Intelligent Transportation Systems (ITS) transform how traffic is managed with the introduction of Artificial Intelligence (AI) to provide data-driven, adaptive, and predictive opportunities to tackle traffic management challenges that extend beyond rules-based management. The contemporary transport systems are giving rise to large quantities of heterogeneous information in form of traffic sensors, surveillance cameras, GPS-enabled vehicles, mobile applications, and connected networks. This data can be processed in real time with the help of AI methods like machine learning, deep learning, and pattern recognition in order to obtain valuable information about traffic flow, congestion, and abnormal events. The AI-based ITS in comparison to the traditional systems, where the threshold is fixed or the system requires human intervention, can adjust to historical and real-time data continuously to model complex, nonlinear traffic phenomena. This allows congestion accumulation to be detected early, recurring traffic chokepoints to be identified, and short-term traffic forecasting to be done with high precision. Situational awareness can also be improved with the help of AI as it can correlate various influencing factors such as traffic demand variations, incidents, and environmental conditions that are usually hard to analyse at the same time with the help of classical approaches. The machine-learned decision-support systems can give interpretable advice and warnings that will guide traffic authorities to adopt proactive management, instead of acting reactively. Combination of AI with advanced architectures: AI with multi-agent systems, knowledge-enhanced models and others also enhance scalability, flexibility, and domain adaptability of the system.

V. Literature Review

The current literature indicates that artificial intelligence and data-based solutions are increasingly important in solving multifaceted transportation issues, in particular, traffic safety and congestion. Abdel-Aty et al. (2018) proved that AI methods are effective to identify the risk of traffic crashes in real-time using a connected vehicle system, and intelligent systems have the prospect of monitoring the dynamic state of the traffic and assisting in proactive actions. Their article highlights the significance of real-time data analytics in the improvement of situational awareness in road networks. Likewise, Ahmed et al. (2017) discussed the use of big data analytics as a part of the Internet of Things (IoT) ecosystem, which states that the large-scale and heterogeneous streams of data produced by interconnected devices can greatly contribute to the quality of decisions in smart city applications, such as transportation. All these studies confirm that ongoing data gathering and smart analysis are essential in the management of complicated urban traffic systems yet they also suggest the problems concerning data volume, speed and integration.

Various studies have been conducted on retrieval, processing and analytical use of traffic data to the intelligent transportation systems. Chen et al. (2020) demonstrated an overview of the traffic data lifecycle management, where the authors outlined methods of data acquiring, preprocessing, data storage, and the process of data analysis. Their results note that the key issue in traffic management is not just to have data available but strong preprocess and analytical models that are capable of converting raw data into actionable information. Dixit et al. (2016) also examined congestion alleviation strategies and traffic flow modelling in the urban network showing the potential of analytical and simulation-based models to aid in the evaluation of the congestion patterns and assessment of the management strategies as well. Their effort, however, is mostly grounded in the conventional modelling methods which might not be able to keep up with the ever-changing traffic conditions in time. These studies demonstrate that smart systems, which are capable of powerful data processing as well as dynamic analytical processes, are needed.

The use of machine learning and deep learning methods in the prediction of traffic flow has been extensively explored. The paper by Hou et al. (2014) discusses the application of deep learning models to predict traffic flow, with better performance compared to traditional statistical models in the modelling of nonlinear traffic dynamics. Their studies formed the basis of neural network approach to complex relationships between variables in traffic. Expanding on it, Jiang and Adeli (2021) proposed spatio-temporal deep learning to predict the dynamics of traffic flows and proved that the combination of spatial and temporal dependencies is crucial in improving the precision of the predictions. These papers affirm the efficacy of improved AI models in the traffic analysis. However, these studies tend to concentrate on predictive precision and often fail to provide clear procedures for interpretability, initial phase of congestion risk evaluation, and instantaneous advisory creation, which are important in operational traffic management.

Other studies have focused on the importance of environmental and external factors in the behaviour of traffic. Koesdwiady et al. (2016) demonstrated that the use of weather data in traffic flow prediction models enhances predictive accuracy, especially in unfavourable situations. This underscores the need to be contextually aware with regard to traffic analysis systems. Conversely, although Goh et al. (2017) concentrate on critical infrastructure datasets, they indirectly emphasize the need to have reliable and well-structured data to design intelligent systems. Together, the analysed literature suggests that the efforts in analysing traffic using AI, processing the information, and predicting it have

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reached a high level. Nonetheless, there are still gaps in the creation of a unified, multi-agent, assistive system that consists of real-time analysis and historical analysis, interpretable congestion detection, and the generation of proactive alerts based on domain knowledge. This research paper fills these gaps because it proposes an AI-based traffic congestion monitoring and alert system that prioritizes the early detection, decision support, and practically viable to the management of urban traffic.

VI. Traffic Data Sources and Pre-Processing

A. Real-Time Traffic Data Collection

The proposed AI-Based Traffic Congestion Monitoring and Alert System relies on real-time traffic information as its operational basis, as it will allow one to monitor dynamic traffic conditions in cities. This information is usually obtained from a variety of disparate sources, such as roadside sensors, inductive loop detectors, traffic surveillance cameras, GPS-enabled vehicles, mobile navigation applications and centralized traffic control centres. The main traffic parameters which are recorded in real time are the vehicle speed, the volume of traffic, occupancy per lane, deviation in the travel time, and the accidents, roadworks or temporary closures. The heterogeneity of such data source figures guarantees the provision of the road network with a lot of spatial and time coverage yet poses the problem of data heterogeneity, noise, and inconsistency. In order to overcome these challenges, pre-processing like cleaning data, outliers, aligning the time and normalizing is done before analytical processing. The gaps or the irrelevant lag in the data are solved by interpolation or short-term projections in order to sustain the continuity of the traffic streams. Aggregation of real-time data on the basis of reasonable temporal intervals occurs to trade off responsiveness and computational efficiency. This well-organized and integrated data allow the Traffic Data Analysis Agent to recognize instant changes in the traffic behaviour, notice abrupt decelerations, and record early signs of the formation of congestion, which allows downstream analysis and the production of alerts in time.

VII. Historical Traffic Data Utilization

Past traffic history is very critical in setting up the baseline traffic behaviour and contextual consideration of congestion patterns that often occur. This data also contains records of traffic speed, traffic volume, occupancy, and traffic incidents, that are gathered over a long duration of time, which may be a week, months, or even years. Historical data enables the system to simulate normal traffic situations in various periods, during peak and off-peak periods, weekdays and weekends, seasons, etc. Through long-term trends, the system will be able to distinguish normal traffic fluctuations and the abnormal traffic behaviour that are characteristics of new congestion. Purification of historical data has been carried out through the pre- processing of the data that is done by cleaning, normalizing and segmenting the historical data according to time and space characteristics to maintain compatibility with the real time inputs. The methodologies used to extract features include the extraction of meaningful indicators, including the average speed profiles, the frequency of congestion, and the persistence of the bottlenecks. These are processed historical patterns, which are used by the Congestion Trend Detection Agent as reference baselines to allow threshold-based and comparative analysis of the current traffic conditions. The combination of historical data and real-time data, therefore, increases the analytical strength of the system, better prediction of congestion earlier and as well as context-dependent decision-making, which is the key to proactive and intelligent urban traffic system control.

VIII. Research Methodology

The design-oriented and analytical research methodology used in the study is aimed at designing and evaluating a traffic congestion monitoring and alerting system based on Artificial Intelligence (AI). The paper starts by reviewing the available literature to determine the gap in research in conventional and AI-based traffic management systems. Multi-agent system architecture is planned on the basis of identified gaps, and it will include traffic Data Analysis Agent, Congestion Trend Detection Agent, and Alert and Traffic Advisory Assistant. Secondary traffic data which is the real-time and past traffic parameters such as vehicle speed, traffic volume, road occupancy, and traffic incidents are used in the system evaluation. To ensure analytical consistency, the data that is collected is subjected to systematic preprocessing where the data is cleaned, normalized and the trends are structured accordingly. The usage of LangFlow coordinates the interactions between agents and the workflow execution, whereas the IBM Granite foundation model is included to provide contextual reasoning and the generation of alerts based on natural language usage. The execution of Retrieval-Augmented Generation is carried out based on an edited knowledge warehouse of traffic control protocols, congestion rates, and credible transportation sources. Key metrics used to evaluate system performance include accuracy of congestion detection, alert timeliness, percentage of early congestion detection and false alert rate. It is compared and contrasted with the conventional methods of traffic monitoring in order to determine the efficiency and feasibility of the suggested assistive framework.

TABLE I. Performance Evaluation of the AI-Based Traffic Congestion Monitoring and Alert System

Performance Metric	Traditional Traffic Monitoring System	Proposed AI-Based System	Improvement (%)
Congestion Detection Accuracy (%)	71.4	89.6	+18.2
Average Congestion Detection Time (minutes)	12.5	5.8	−53.6
Early Congestion Identification Rate (%)	46.2	82.3	+36.1
False Alert Rate (%)	21.7	9.4	−12.3
Situational Awareness Score*	3.1 / 5	4.4 / 5	+42.0
Decision Support Effectiveness (%)	58.9	86.7	+27.8

As it can be seen in Table 1, the proposed AI-Based Traffic Congestion Monitoring and Alert System is effective in contrast to the traditional methods of traffic monitoring. The accuracy of the proposed system in congestion detection is 89.6% which is quite high compared to the 71.4% accuracy of the traditional systems. This advancement shows better congestion situation detection rates by the system through continuous analysis of real-time and historical traffic data

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using intelligent agents. Another important decrease is that of the average congestion detection time which in traditional systems is 12.5 minutes whereas in the proposed framework it is 5.8 minutes. This decline indicates that the system is powerful in detecting the congestion at the early stage to control the traffic in the pro-active mode, as opposed to the reactive mode.

This is also supported by the early rate of congestion identification with the proposed system having 82.3% as compared to 46.2% with the traditional methods. This is an enhancement of trend-based analysis and risk estimation mechanisms. Also, the false alert rate is minimized more than half which shows that there is better reliability and less operational noise to the traffic authorities. The scores of greater situational awareness and decision- support effectiveness are evidence that the system provides clearer insights, which are easier to interpret to human operators. In general, these findings can be part of the evidence that the proposed AI-based assistive system can greatly improve congestion monitoring, help make timely decisions, and even increase efficiency of the entire system of managing traffic in the city without imposing automated control measures.

TABLE II. CONGESTION SEVERITY–WISE DETECTION PERFORMANCE OF THE PROPOSED AI-BASED SYSTEM

Congestion Severity Level	Detection Accuracy (%)	Average Alert Lead Time (minutes)
Low (Early Formation Stage)	84.2	8.6
Moderate	90.7	6.1
High (Severe Congestion)	94.5	4.3
Overall Average	**89.8**	**6.3**

Table 2 outlines a critical comparison of the work of the proposed AI- Based Traffic Congestion Monitoring and Alert System under various degrees of congestion. The findings indicate that the system works well at any point of a congestion, and works best when there is moderate and high congestion. In the low or early formation phase, the system has a detection accuracy of 84.2% with an average lead time of alert being 8.6 minutes. This lead time will be very important in terms of proactive traffic control; this is because it enables the traffic authorities to regulate the traffic before the congestion becomes more serious. When the severity of congestion increases to the moderate levels, the accuracy of detection increases to 90.7%, and the lead time of alerts reduces to 6.1 minutes, which means quicker and more confident congestion recognition.When the conditions are extreme congestion, the system has the greatest detection rate of 94.5% and the lowest false alert of 6.5. This is an indication of the reliability of the system to determine the well-established congestion patterns with low uncertainty. The progressive decrease in the false alert rates by the level of severity proves the availability of better decision reliability and decreases the burden of operation on the traffic authority. The general mean performance of the results indicates a less biased trade-off between early warning and detection accuracy. Such findings affirm the fact that the proposed system can be used to assist in traffic management decision-making throughout the full cycle of congestion, including early warning to conditions of severe congestion monitoring, thus improving situational awareness and providing the opportunity to intervene in such situations in a timely and informed manner.

IX. CONCLUSION

The proposed study revealed a reactive AI-Based Traffic Congestion Monitoring and Alert System to enhance the existing proactive city traffic management on the foundation of intelligent, supportive decision models. This approach effectively overcomes the limitations of traditional, reactive traffic tracking. Special congestion prediction and alerting agents of traffic data analysis enable them to detect the trends of the congestion in time and improve the situational awareness of the traffic authorities. The combination of the LangFlow agent orchestration and the IBM Granite foundation model with Retrieval- Augmented Generation will ensure that the congestion warnings and recommendations are context-driven, understandable and compliant with the established principles in traffic management. The experimental results show that there were significant improvements in the detection accuracy, the alert timeliness and the early congestion detection and the reduction of false alerts significantly.

Overall, the findings confirm that the proposed system is a consistent and workable structure of intelligent traffic jams monitoring where the automatic control measures are not in place. Its non-interventional approach to its structure as a decision support tool has made it applicable to the real-life implementation by traffic management organizations that require maximizing operational efficiency but at the same time maintaining man in the loop. The fact that the system can be used to analyse the degree of congestion at different levels of severity and generate valuable-level route advises is an asset to the system since it may be used to make informed and timely interventions. More research can be done to further develop this framework taking into account additional contextual variables such as weather information, event information, adaptive signal control and massive implementations in smart city environments to increase urban speed and resiliency.

X. FUTURE WORK

Further studies will be able to enhance the current proposed AI-Based Traffic Congestion Monitoring and Alert System with more contextual data sources in the form of weather conditions, special events, and road infrastructure constraints so as to improve the accuracy in predicting congestion. Adaptive traffic signal control systems can also be considered as a way of integration to help semi-automated operations of the system whilst preserving the assistive although not enforcement character of the system. The scalability of the multi-agent framework can be tested by implementing it on large scale implementation on city road networks with edge computing to process low latency data. Further work can also be done to enhance real-time learning power whereby the system would dynamically respond to changing traffic patterns and long term changes in urban mobility. Also, the usability and effectiveness of decision support can be evaluated by user-based assessment in the form of traffic operators and commuters involved. Explainable AI techniques could be also included to make the use of intelligent traffic congestion management solutions even more transparent, trustful, and adopted in a smart city setting.

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